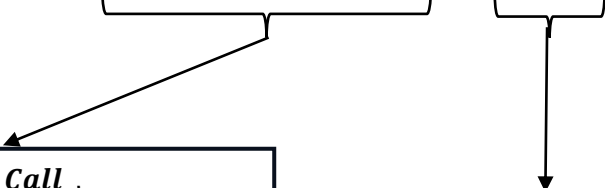


Fibonacci Series Time Complexity with Backward Substitution Method

***In fibonacci series : $T(n) = T(n - 1) + T(n - 2)$, where `n` is greater than 1 ,
when `n` = 0 , $T(n) = 1$ and $T(n) = 1$, when `n` = 1, these are base cases for
fibonacci series.***

To start with: –

***$T(n) = T(n - 1) + T(n - 2) + 1$, here 1 is for adding two
numberd returns from $T(n - 1), T(n - 2)$.***

$$T(n) = T(n - 1) + T(n - 2) + C \text{ or } 1$$


***Recursive Call ,
Problem Size : $(n - 1), (n - 2)$,
Means: Times required
to solve the remaining
problem sizes generated
from: $N - 1, N - 2$.***

***Add, the two returned numbers ,
from: $T(n - 1), T(n - 2)$.***

$$T(n) = \begin{cases} 1 & , \text{if } n = 0 \\ 1 & , \text{if } n = 1 \\ T(n-1) + T(n-2) + \text{Cor1} & , \text{if } n > 1 \end{cases}$$

Assuming, $T(n-1) \approx T(n-2)$, we get :

$$T(n) = T(n-1) + T(n-1) + 1$$

$$T(n) = 2 \times T(n-1) + 1$$

By Backward Substitution:

Substitution the values of $T(n-1)$ in the recurrence equation, one gets the following equations:

$$= 2 \times [2 \times T(n-1-1) + 1] + 1$$

$$= 4T(n-2) + 3$$

Substitution the values of $T(n-2)$ in the recurrence equation, one gets the following equations:

$$= 2[4T(n-2-1) + 3] + 1$$

$$= 8T(n-3) + 7$$

Substitution the values of $T(n - 3)$ in the recurrence equation, one gets the following equations:

$$= 2[8T(n - 3 - 1) + 7] + 1$$

$$= 16T(n - 4) + 15$$

Hence by repeating the process, one can observe that at the i th iteration, this equation would be as follows:

$$T(i) = 2^i T(n - i) + 2^i - 1$$

Lets say the Fibonacci series, ends up on base case: $T(1)$, then, $i = n - 1$. As $T(1) = 1$ [base case] $\therefore n - i = 1 \Rightarrow i = n - 1$.

If $i = n - 1$, we will have:

$$T(i) = 2^{n-1} T(n - (n - 1)) + 2^{n-1} - 1$$

$$= 2^{n-1} T(n - n + 1) + 2^{n-1} - 1$$

$$= 2^{n-1} T(1) + 2^{n-1} - 1$$

$$= 2^{n-1} \times 1 + 2^{n-1} - 1 \text{ [We know, } T(1) = 1 \text{]}$$

$$= 2^{n-1} + 2^{n-1} - 1$$

$$= 2 \times (2^{n-1}) - 1 \quad [\text{As, } 2^{n-1} + 2^{n-1} = 2 \times (2^{n-1})]$$

$$= 2^{n-1+1} - 1$$

$$= 2^n - 1$$

$$= O(2^n - 1)$$

$$= O(2^n)$$

Therefore, $O(2^n)$ is the answer.
